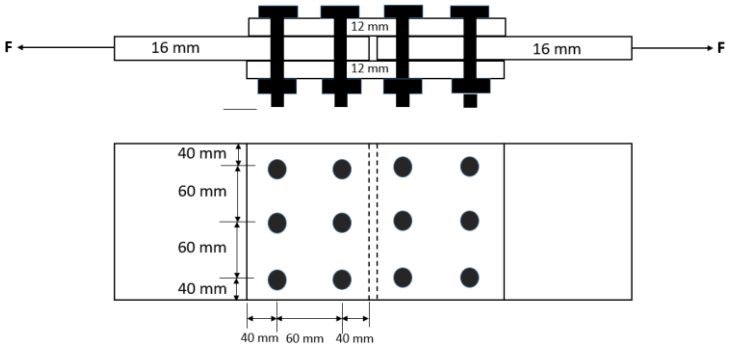
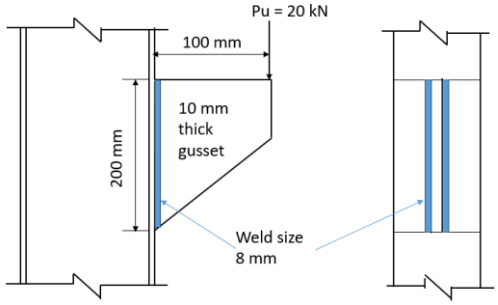
MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING			
MAKE-UP EXAMINATION			
Level	UG	**Curriculum**	2022
Program Name	Civil engineering	**Course Type**	C
Course Name	Basic Structural Steel Design	**Course Code**	CIE 3121
Semester	5	**Max Marks**	50
Date		**Duration**	3 Hrs

Instructions:

- **Answer ALL the questions.**
- **Missing data if any may be suitably assumed.**
- **Assume Fe410 steel with f_y = 250 N/mm^2 unless specified otherwise.**
- **Use of IS800: 2007 and SP6 is permitted.**

Q. No.	Question	Marks	CO	PO	BL
1A	Determine the maximum load carrying capacity of a butt connection shown in the figure. The bolts are M16 of grade 4.8 	5	1	1,2, 6,8, 10, 12	3
1B	Determine the adequacy of the welded connection shown in figure. Assume field/site weld. 	5	1	1,2, 6,8, 10, 12	3
2A	Determine the size of a suitable equal angle section to carry a factored tensile load of 200 kN. Also determine the number of bolts required for the connection. Assume M20	5	2	1,2, 6,8,	3

	bolts of grade 4.6 for the connection. Also, assume a 16 mm thick gusset. Adequacy check need not be carried out.			10, 12	
2B	Determine the compression capacity of 2, ISA 55×55×10 connected back to back to a gusset plate of 12 mm thick. The section is of 2.5 m length and connected to the gusset using 2 bolts (k=0.85).	5	3	1,2, 6,8, 10, 12	3
3A	Design a column of length 3.5 m to carry a load of 1000 kN. At one end the column is retrained against translation and rotation, at other end it is restrained against translation but free to rotate. Assume f_{cd} = 135 N/mm^2 for initial area calculation.	5	3	1,2, 6,8, 10, 12	3
3B	A column ISWB 200 @ 28.8 kg/m carrying a factored load of 1900kN is to be supported on a concrete footing. Design a suitable slab base plate. The grade of the concrete pedestal is M25. Connections need not be designed.	5	4	1,2, 6,8, 10, 12	3
4A	Explain the main failure modes of hot-rolled beams	5	5	1,2, 6,8, 10, 12	2
4B	Design a laterally unsupported beam to carry a factored UDL of 15 kN/m resting over a simply supported span of 3.5 m. Assume ends of the columns are restrained against twisting and warping. Check for shear and deflection need not be carried out. Assume f_{bd} = 100 N/mm^2 for the initial design approximation. Check for deflection need not be carried out.	5	5	1,2, 6,8, 10, 12	3
5A	Determine the bending strength of laterally supported section ISWB 600 @ 145.1 kg/m used as a simply supported beam of span 5 m. Assume V_u = 0.9 V_d.	5	5	1,2, 6,8, 10, 12	3
5B	Check ISMB 250 @ 37.3 kg/m for web buckling and crippling strength near supports. The stiff bearing is over a length of 125 mm. The factored shear force on the beam is 100 kN.	5	5	1,2, 6,8, 10, 12	3

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	10
2	5
3	10
4	5
5	20

BL	Total Marks
2	5
3	45
4	-
5	-